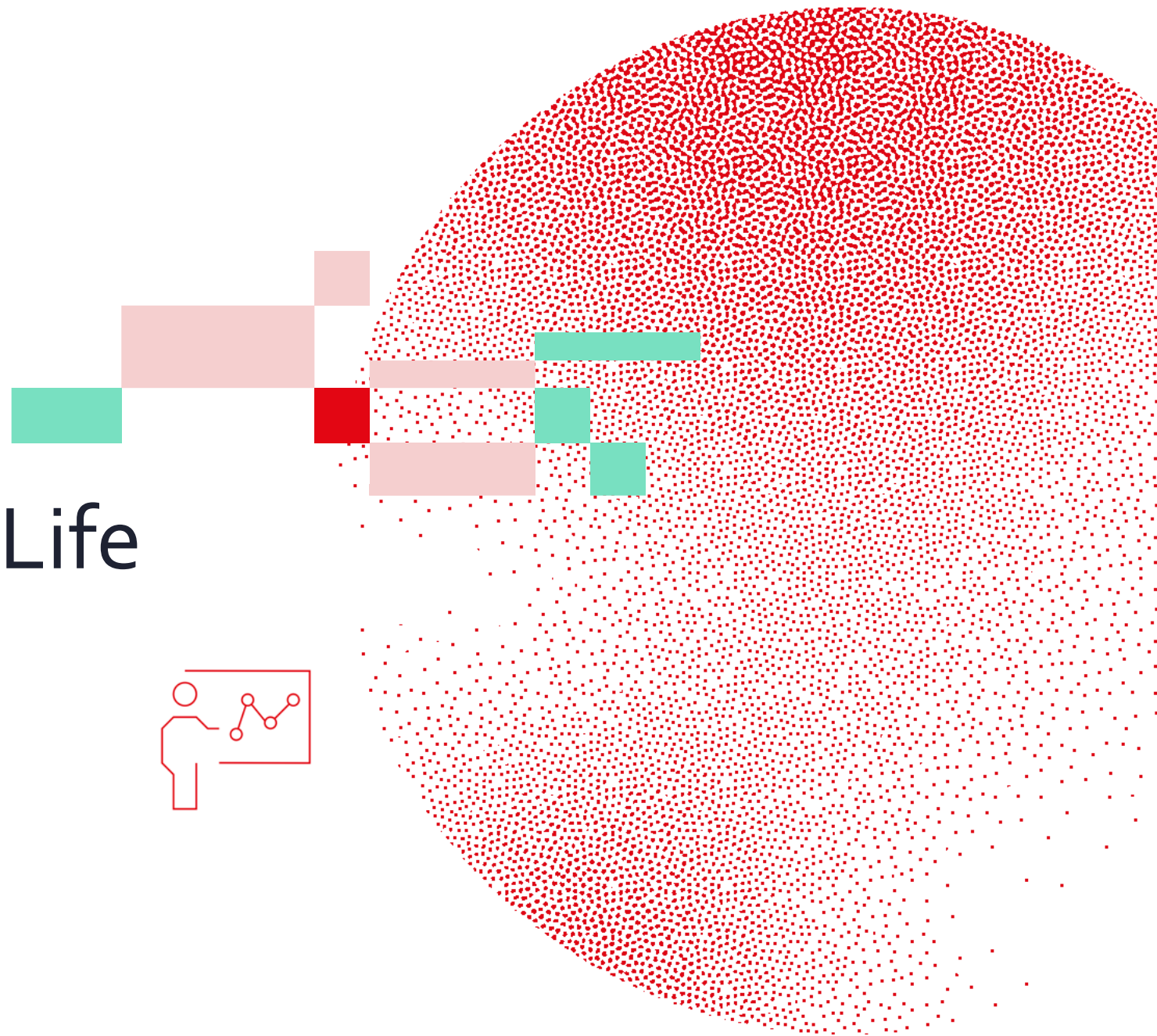




Swiss Institute of
Bioinformatics

First Steps with R in Life Sciences: Statistics

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Statistical hypothesis testing

Why do we need a statistical test?

Biological measurements are noisy: two groups never have exactly the same mean, even when nothing really differs between them. A test helps us decide whether a difference is real or could simply be due to chance.

Two hypotheses in competition

- **H₀, the null hypothesis:** "nothing is going on" (e.g. no difference between groups)
- **H₁, the alternative hypothesis:** what we actually want to show

Example

- H₀: KO and WT mice have, on average, the same body weight
- H₁: the average body weight of KO mice is different from that of WT mice

The p-value in plain words

Imagine that H_0 is true (e.g. KO and WT mice really have the same weight)
It would mean that genotype doesn't matter for weight

If it doesn't, the KO/WT labels are meaningless: we could shuffle them randomly.

p-value = how often a random shuffle of the labels gives a difference as big as the real one

$p = 0.001 \rightarrow$ shuffling almost never does this $\rightarrow$ the labels matter

$p = 0.4 \rightarrow$ shuffling does this often $\rightarrow$ can't tell whether labels matter ($\neq$ proof they don't)

The usual recipe

- Before looking at the data, choose a significance level, traditionally **0.05**
- Run the test in R: it gives you a test statistic and a p-value
- $p\text{-value} < 0.05 \rightarrow$ the difference is called "statistically significant"

A significant result means the difference is probably not due to chance. It does not tell you that the difference is large or biologically important.

t-test

Goal

- Compare the mean of a continuous measurement between **two groups**
- Is the difference between the two means larger than what we would expect by chance?

Life-science examples

- Body weight of KO vs WT mice
- Expression of a gene in treated vs untreated cells
- Cholesterol level of patients on diet A vs diet B

Assumptions

- Observations are independent (e.g. each mouse is measured only once)
- In each group, the values roughly follow a normal ("bell-shaped") distribution

How does the t-test work?

The t-test compares two things:

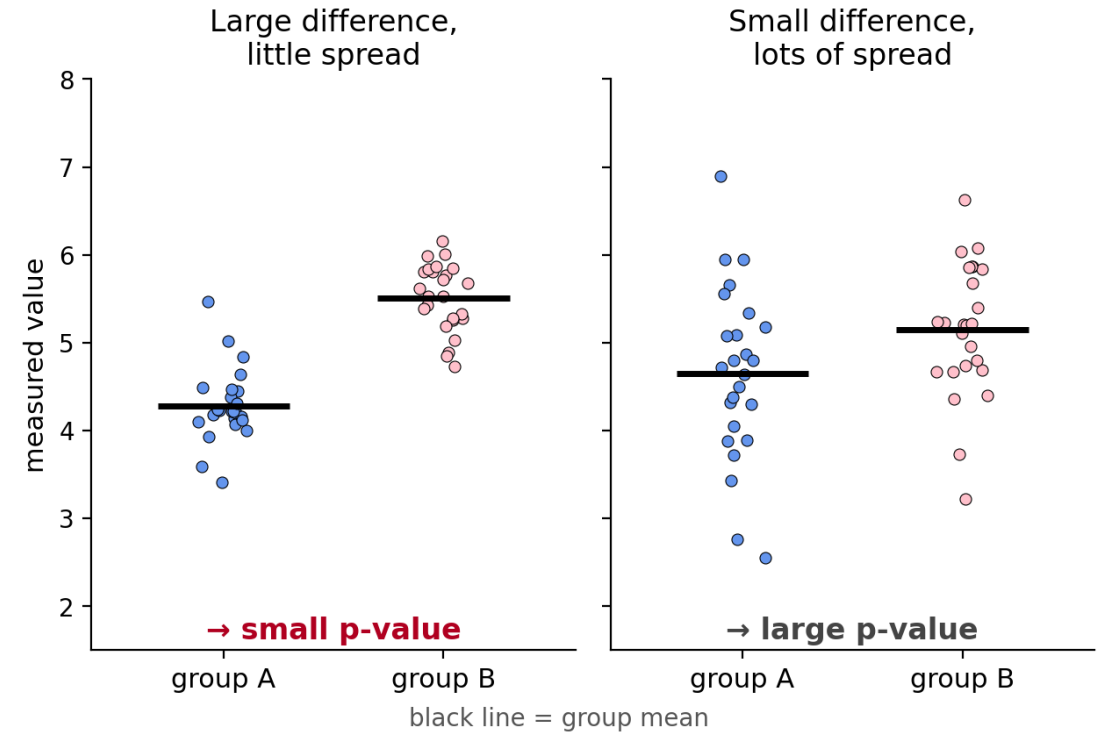
- the **difference between the group means** (the signal)
- the **spread of the values** within each group (the noise)

Both are combined into one number, the **t statistic**:

$t \approx \text{difference of means} / \text{its uncertainty}$

- Large difference, little spread $\rightarrow$ large $t \rightarrow$ small p-value
- Small difference or lots of spread $\rightarrow$ small $t \rightarrow$ large p-value

More samples per group = less uncertainty = easier to detect a real difference.



Example data set: iris

Let's compare *Iris virginica* and *Iris versicolor* petal lengths.

```
data(iris)
# we limit the data to 2 species
iris_f = iris[ iris$Species %in% c('versicolor','virginica') , ]
iris_f$Species = factor(iris_f$Species)

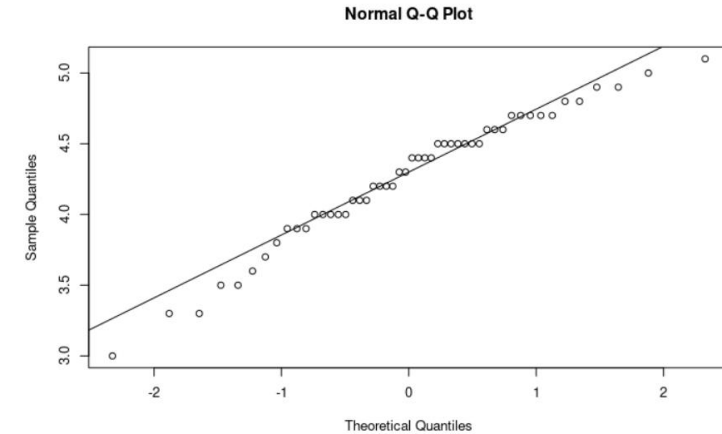
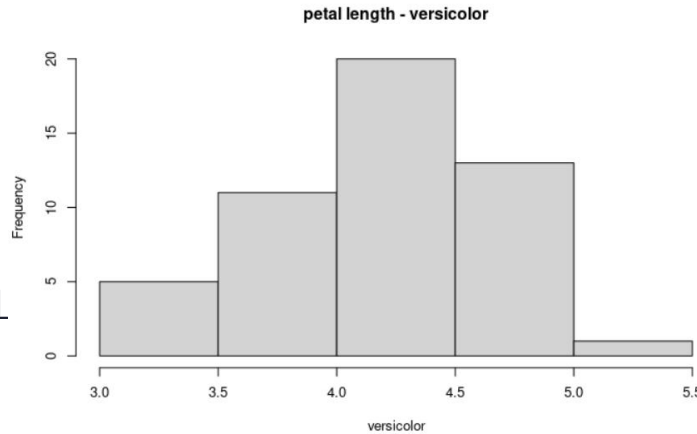
tapply( iris_f$Petal.Length, iris_f$Species, mean )
versicolor  virginica
      4.260      5.552

plen_versicolor = iris_f$Petal.Length[iris_f$Species=='versicolor']
plen_virginica   = iris_f$Petal.Length[iris_f$Species=='virginica']
```

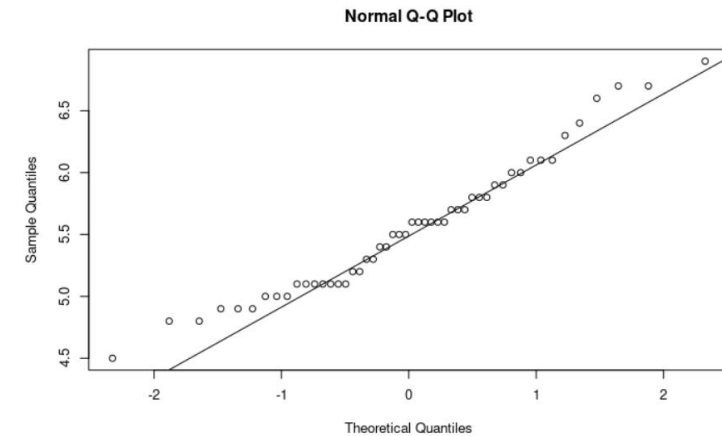
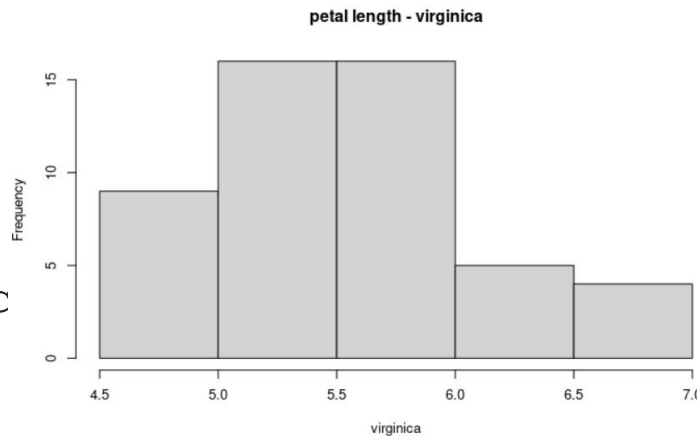
Check normality of the data with plots

We use histograms and QQplot to visually assess normality

```
par(mfrow=c(2,2))  
hist(plen_versicolor,  
     xlab="versicolor",  
     main="petal length - versicol  
qqnorm(plen_versicolor)  
qqline(plen_versicolor)
```



```
hist(plen_virginica,  
     xlab="virginica",  
     main="petal length - virginic  
qqnorm(plen_virginica)  
qqline(plen_virginica)  
dev.off()
```



Check normality of the data with tests

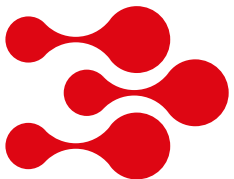
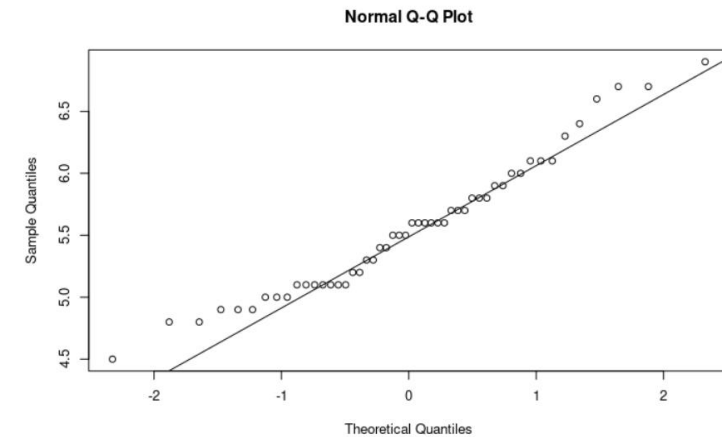
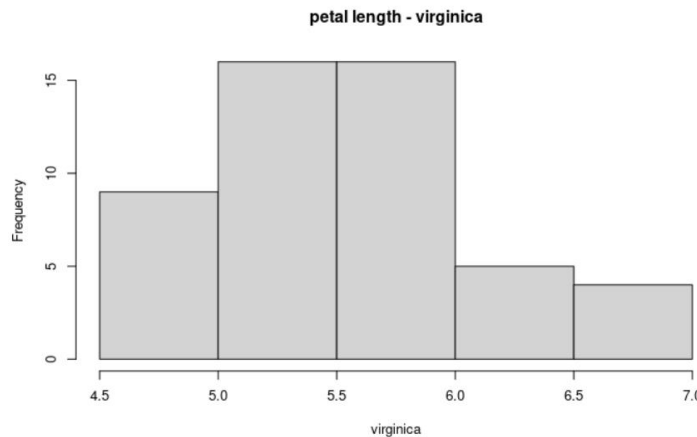
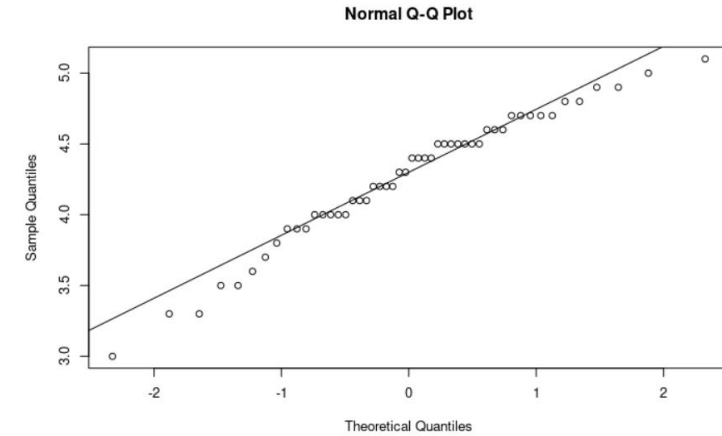
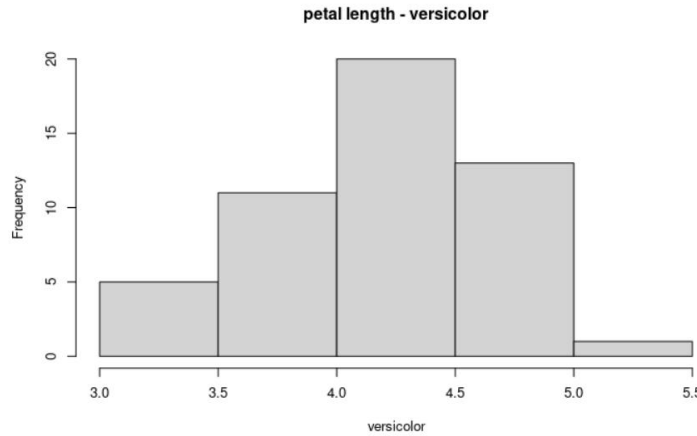
The **Shapiro-Wilk test** checks normality.

Here H_0 = "the data are normal", so:

- $p\text{-value} > 0.05$: no evidence against normality
- $p\text{-value} < 0.05$: the data are probably not normal

```
shapiro.test(plen_versicolor)  
W = 0.966, p-value = 0.1585
```

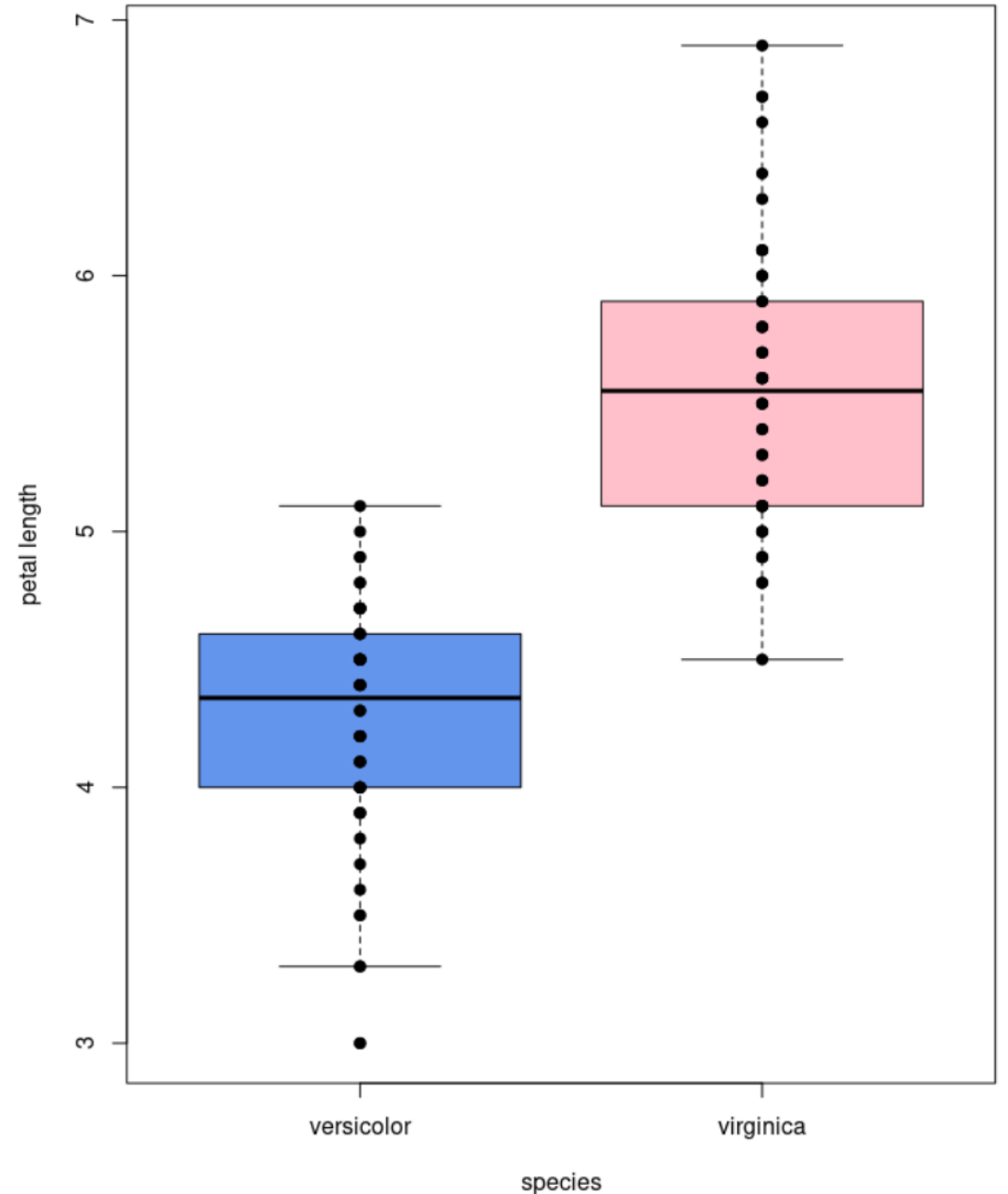
```
shapiro.test(plen_virginica)  
W = 0.96219, p-value = 0.1098
```



The t-test still works well when data are only roughly normal: no need to be too strict.
Always look at the plots, not only at the test!

Visualize group differences

```
boxplot(Petal.Length ~  
Species,  
        data=iris_f,  
        col=c("cornflowerblue"  
              , "pink"),  
        ylab="petal length",  
        xlab="species")  
  
points(Petal.Length ~  
Species,  
        data=iris_f,  
        col="black",  
        pch = 19)
```



Function t.test()

```
t.test(plen_versicolor, plen_virginica)
```

```
t.test(Petal.Length ~ Species, data=iris_f) #equivalent to the above
```

Welch Two Sample t-test

```
data:  Petal.Length by Species
```

```
t = -12.604, df = 95.57, p-value < 2.2e-16
```

```
alternative hypothesis: true difference in means between group versicolor and group  
virginica is not equal to 0
```

```
95 percent confidence interval:
```

```
-1.49549 -1.08851
```

```
sample estimates:
```

mean in group versicolor	mean in group virginica
4.260	5.552

Reading the t.test() output

t = -12.604, df = 95.57, p-value < 2.2e-16

- **t**: size of the difference compared to the noise (the sign only says which group is larger)
- **df** (degrees of freedom): linked to the number of observations, you can ignore it for now
- **p-value**: < 2.2e-16 is the smallest value R displays, so extremely small → we reject H_0

95 percent confidence interval: -1.49549 -1.08851

- Plausible range for the true difference of means (versicolor – virginica). It does not contain 0, which agrees with the small p-value.

mean in group versicolor	mean in group virginica
4.260	5.552

- The two group means

How to report it: "On average, *I. virginica* petals are 1.3 cm longer than *I. versicolor* petals (Welch t-test, $p < 0.001$)."

t.test object

- `t.test()` and other tests return a **list** that can be assigned to a variable.
- View the names of the list's slots using **names()**.
- Access the elements of a list using the **\$** or the **[[]]** operators.

```
test_res = t.test(Petal.Length ~ Species, data=iris_f)
```

```
names(test_res)
```

```
[1] "statistic"    "parameter"    "p.value"      "conf.int"     "estimate"  
[6] "null.value"   "stderr"       "alternative"  "method"       "data.name"
```

```
test_res[['p.value']]
```

```
[1] 4.900288e-22
```

Paired data

Sometimes both measurements come from the **same individuals** (or from matched pairs). The two groups are then not independent → use the **paired t-test**.

Life-science examples

- Blood pressure of the same patients before and after a treatment
- Tumour vs adjacent normal tissue from the same patient
- The same mice weighed at the start and at the end of a diet

How it works

- Compute the difference within each pair (e.g. after – before)
- Test whether the mean of these differences is different from 0

Assumption

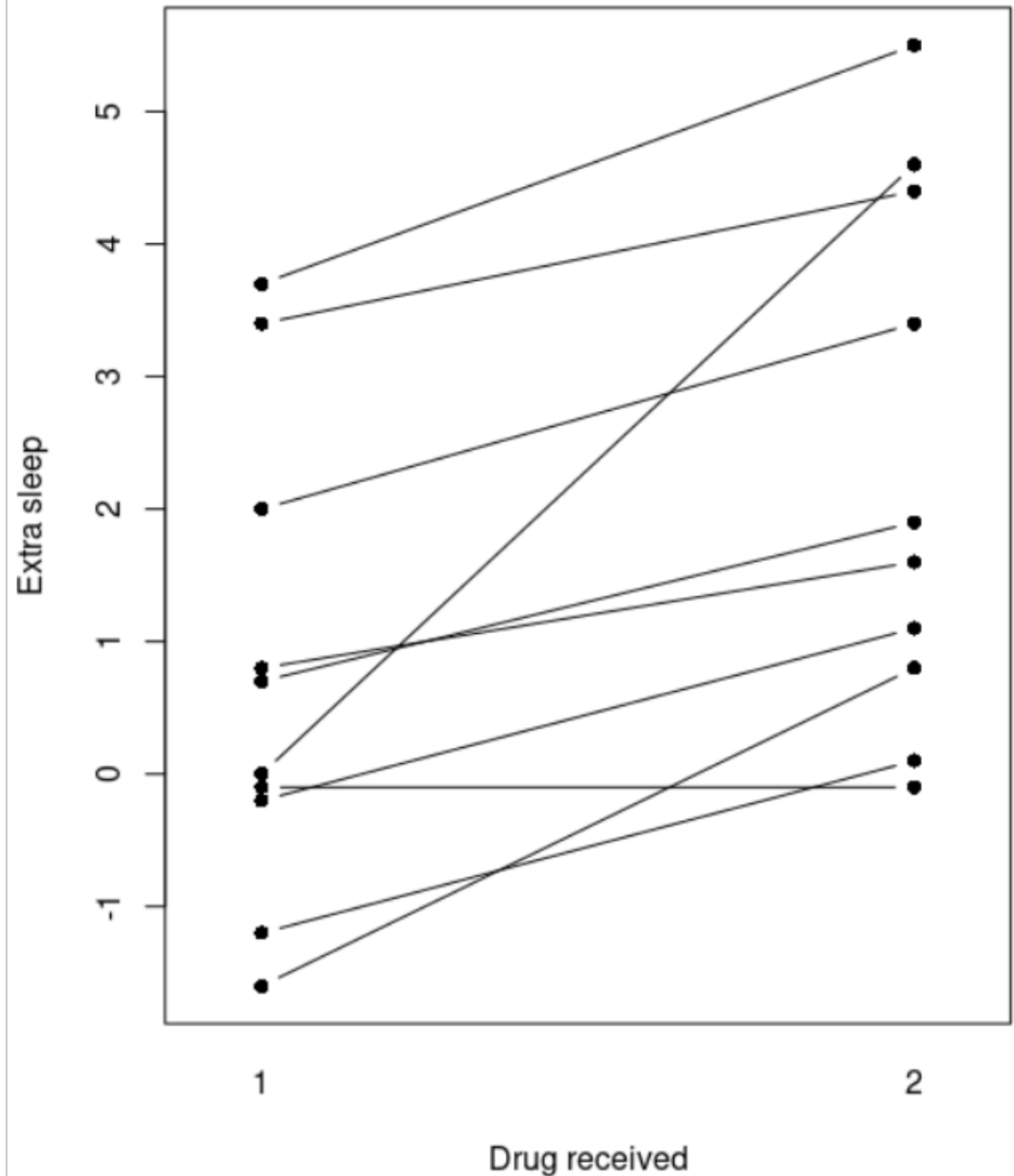
- The differences are roughly normally distributed

Paired data representation

```
data(sleep)
head(sleep, n=3)
```

	extra	group	ID
1	0.7	1	1
2	-1.6	1	2
3	-0.2	1	3

```
interaction.plot(response=sleep$extra,
                 x.factor=sleep$group,
                 trace.factor=sleep$ID,
                 legend=FALSE, type="b",
                 lty=1, pch=16,
                 xlab="Drug received",
                 ylab="Extra sleep")
```

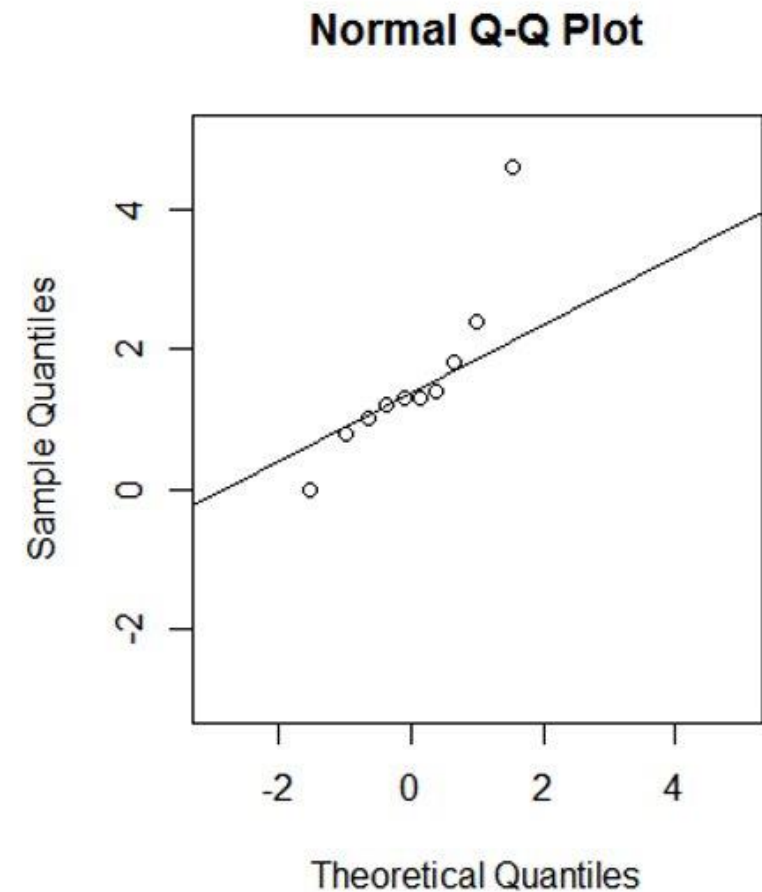
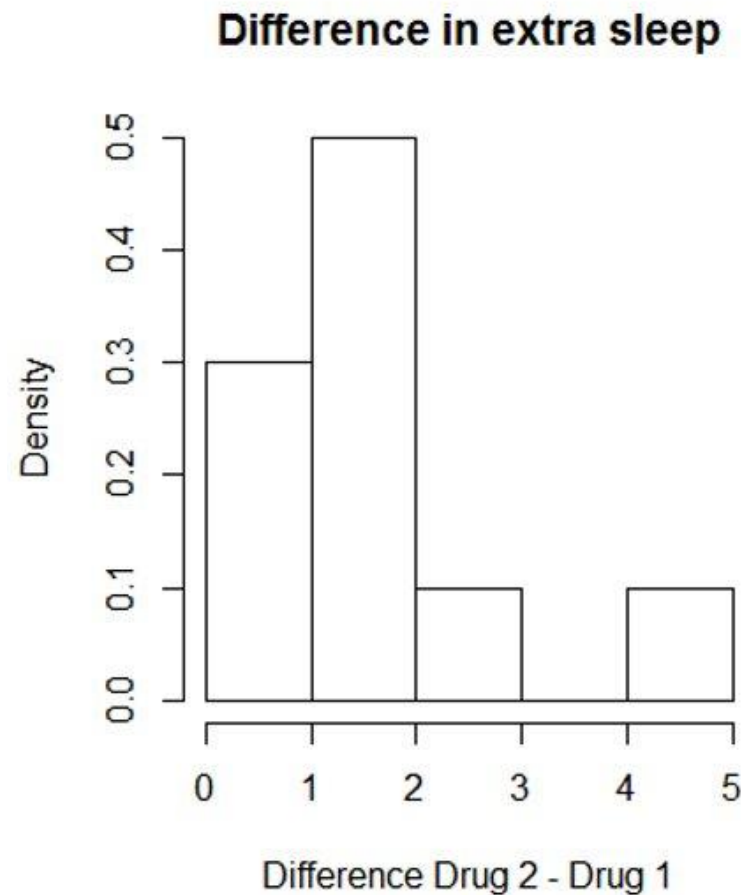


Check normality of the differences between pairs

```
difference = sleep$extra[sleep$group==2]-sleep$extra[sleep$group==1]
```

```
hist(difference, freq=FALSE, xlab="Difference Drug 2 - Drug 1",  
     main="Difference in extra sleep", col="white")
```

```
qqnorm(difference)  
qqline(difference)
```



Paired t-test

```
# using a paired t-test:
```

```
t.test(sleep$extra[sleep$group==1],  
       sleep$extra[sleep$group==2], paired=TRUE)
```

```
# paired values must be at the same position in the two vectors
```

```
# do not use formula notation (extra~sleep) for paired t-test
```

Paired t-test

```
data:  sleep$extra[sleep$group == 1] and sleep$extra[sleep$group == 2]
```

```
t = -4.0621, df = 9, p-value = 0.002833
```

```
alternative hypothesis: true difference in means is not equal to 0
```

```
95 percent confidence interval:
```

```
-2.4598858 -0.7001142
```

```
sample estimates:
```

```
mean of the differences
```

```
-1.58
```

Choosing a t-test, and alternatives

Which t-test should I use?

- Two independent groups (e.g. KO vs WT mice):
`t.test(x, y)`
- The same individuals measured twice (e.g. before / after treatment):
`t.test(x, y, paired=TRUE)`

What if the data are far from normal?

- Small deviations are fine: the t-test is quite robust
- For data that are clearly not normal (very skewed, strong outliers), alternative tests exist. They are called **non-parametric tests** and do not assume any particular distribution.
- They are not covered in this course. If you need them, look at the resources at the end of the course or ask a statistician.

Let's practice - 10

Come back to the mice data-set stored in the "mice_data" data frame.

1. Considering WT mice weight and KO mice weight separately, check the assumption of normality graphically.
2. Make an appropriate plot to visualize the mouse weights grouped by genotype.
3. Perform a test to see whether the mouse weight is different between the two genotypes.
4. *Repeat steps 1 to 3 for the diet variable.*

Bivariate linear correlation

Goal: measure how strongly two continuous variables are linearly related

- Examples: expression of two genes across samples; BMI and blood pressure of patients
- The correlation coefficient goes from -1 (perfect negative) through 0 (none) to +1 (perfect positive)

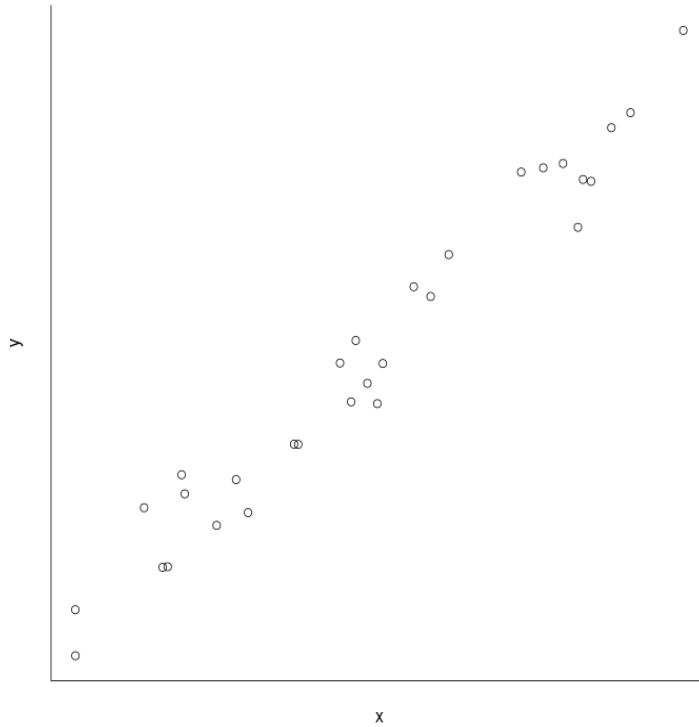
`cor()` computes the correlation between two variables

- **Default:** `method="pearson"` (linear correlation)
- **Other options:** `method="spearman"`, `method="kendall"` (rank-based correlations)

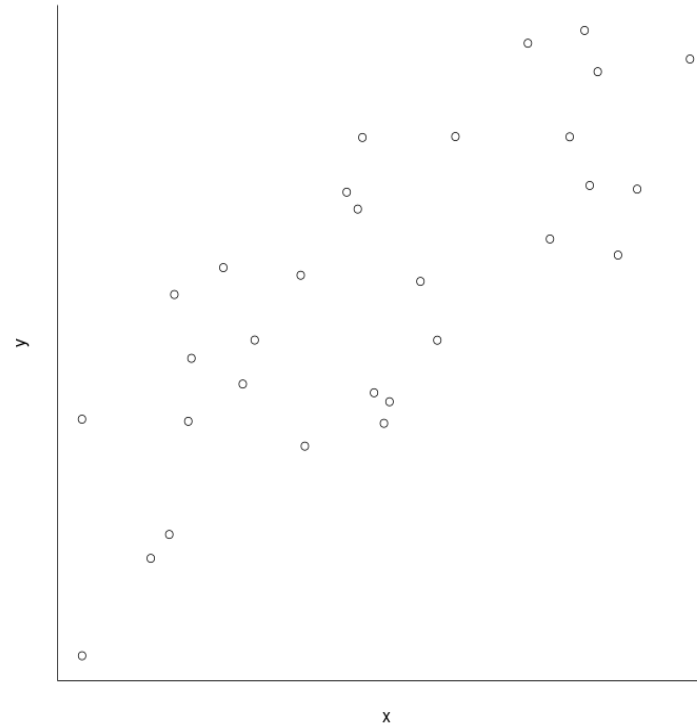
`cor.test()` computes the correlation **and** tests whether it is different from 0 (p-value)

Correlation does not mean causation!

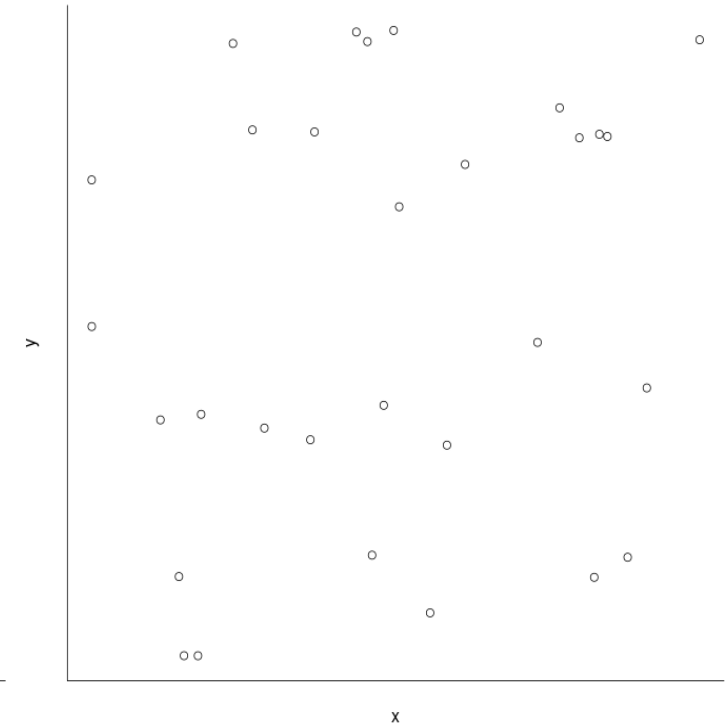
Visualization of linear correlation



Strong linear correlation

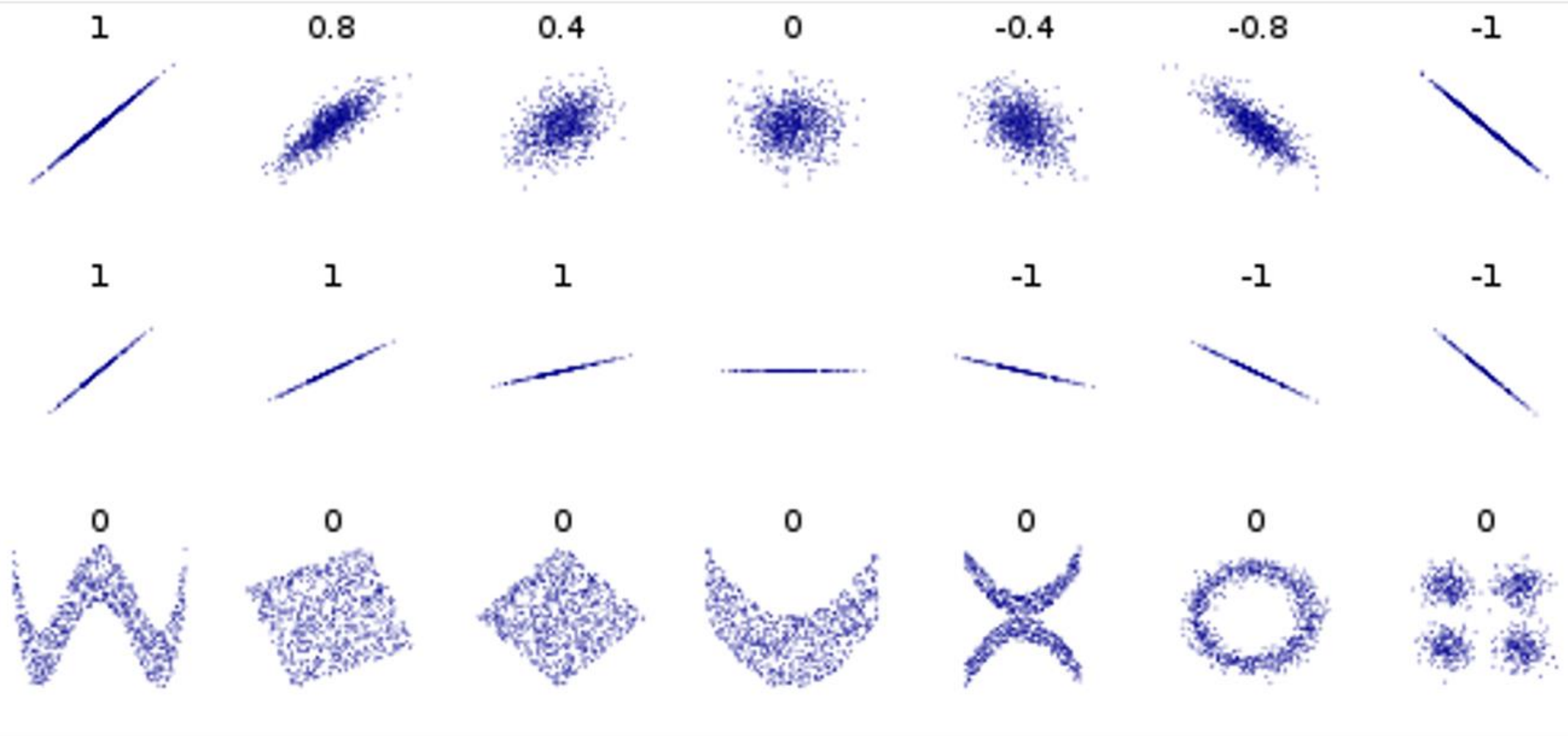


Medium-strong linear correlation



No correlation

More visualization of linear correlation

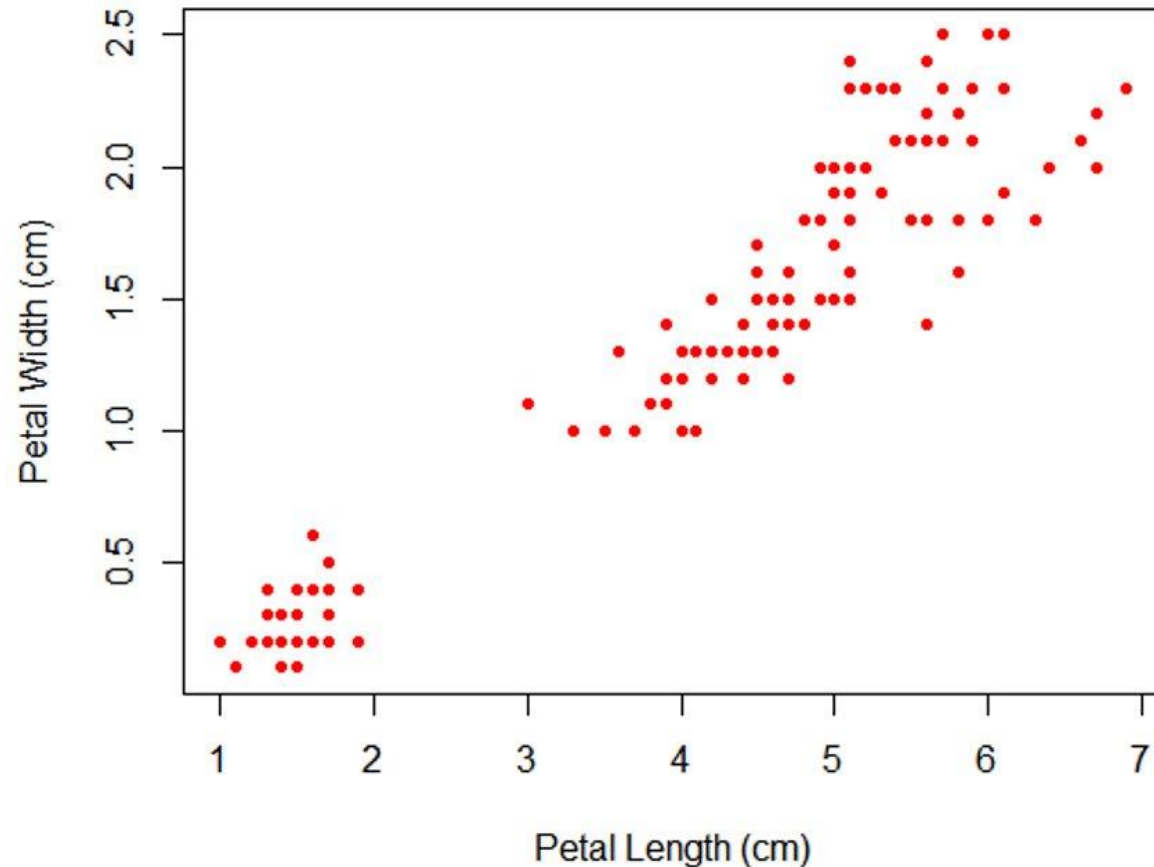


correlation reflects the noisiness and direction of a linear relationship (top row)
but not the slope of that relationship (middle)
nor many aspects of nonlinear relationships (bottom).

Image credit:
wikipedia user DenisBoigelot,
under the CC0 1.0 license

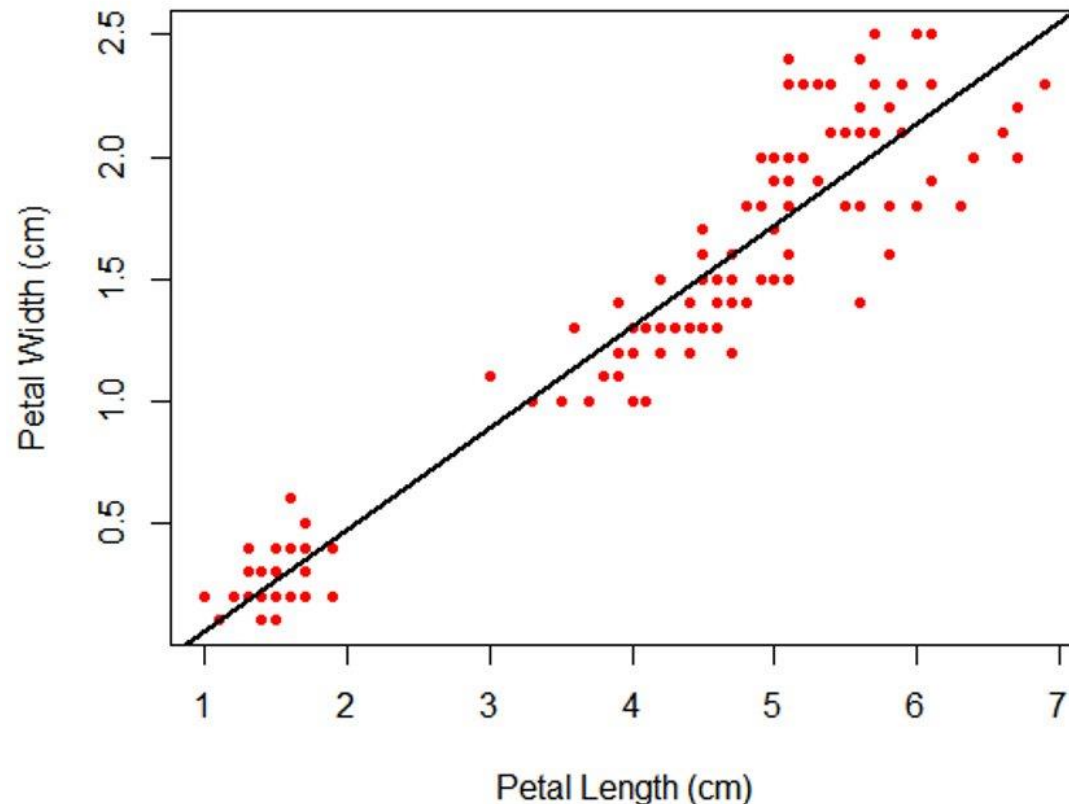
Scatter plot

```
plot(iris$Petal.Length, iris$Petal.Width,  
     col="red", pch=20,  
     xlab="Petal Length", ylab="Petal Width")
```



Scatter plot

```
plot(iris$Petal.Length, iris$Petal.Width,  
     col="red", pch=20,  
     xlab="Petal Length", ylab="Petal Width")  
abline(lm(iris$Petal.Width~iris$Petal.Length),  
       col="black", lwd=2)
```



Linear correlation

```
cor(iris$Petal.Length, iris$Petal.Width, method="pearson")  
[1] 0.9628654
```

```
cor.test(iris$Petal.Length, iris$Petal.Width, method="pearson")
```

Pearson's product-moment correlation

data: iris\$Petal.Length and iris\$Petal.Width

t = **43.387**, df = 148, p-value < **2.2e-16**

alternative hypothesis: true correlation is not equal to 0

95 percent confidence interval:

0.9490525 0.9729853

sample estimates:

cor

0.9628654

Linear regression

Goal

Describe how an **outcome** variable changes with an **explanatory** variable, using a straight line. The line can also be used to make predictions.

Life-science examples

- How does blood glucose change with BMI?
- How does the response to a drug change with the dose?
- How does height change with age in children? (our example)

In R, the model is written as a formula

```
outcome ~ explanatory variable  
Height ~ Age           # "Height explained by Age"
```

Correlation: how strongly are X and Y related?

Regression: how much does Y change when X increases?

Simple linear regression

R finds the straight line that best fits the points:

$$\hat{y} = a + b \times x$$

- **x**: explanatory variable (e.g. Age) **y**: outcome (e.g. Height)
- **a, the intercept**: value of y when x = 0
- **b, the slope**: how much y changes on average when x increases by 1 (e.g. cm per year)
- **Residuals**: vertical distance between each point and the line (what the line does not explain)

Assumptions (checked with plots)

- The relationship looks roughly like a straight line
- Observations are independent
- Residuals have a similar spread all along the line
- Residuals are roughly normally distributed

Summary of the data

CLASS dataset, from the program SAS

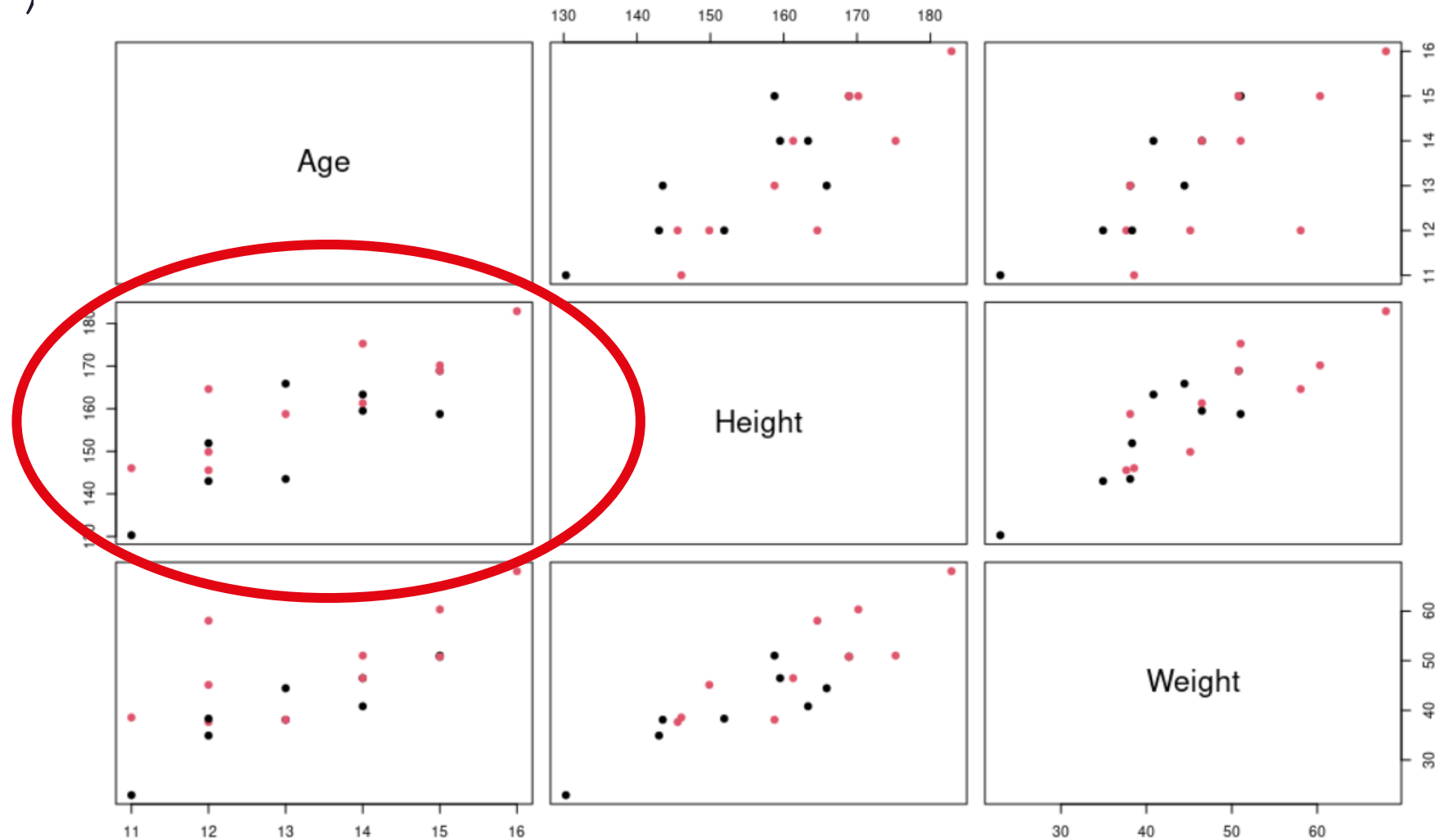
(names removed and units have been modified from imperial to metric)

```
class_data <- read.csv("course_dataset/class.csv")
class_data$Gender=as.factor(class_data$Gender)
summary(class_data)
```

Gender	Age	Height	Weight
F: 9	Min. :11.00	Min. :130.3	Min. :22.91
M:10	1st Qu.:12.00	1st Qu.:148.0	1st Qu.:38.22
	Median :13.00	Median :159.5	Median :45.13
	Mean :13.32	Mean :158.3	Mean :45.37
	3rd Qu.:14.50	3rd Qu.:167.4	3rd Qu.:50.92
	Max. :16.00	Max. :182.9	Max. :68.04

Visual summary of the data

```
pairs(class_data[2:4] , col = class_data$Gender,  
      pch=19 )
```



The `lm()` function

`lm()` fits a linear model and returns an object containing the result:

```
my_model <- lm(outcome ~ explanatory, data =  
my_data)
```

Useful functions to apply to the model

- `my_model` (just print it): the intercept and the slope
- `plot(my_model)`: plots to check the assumptions
- `summary(my_model)`: p-values and R-squared
- `coef(my_model), residuals(my_model)`: extract the coefficients or the residuals

Simple linear regression code

```
model_height_age <- lm(Height~Age, data=class_data)
model_height_age
```

Call:

```
lm(formula = Height ~ Age, data = class_data)
```

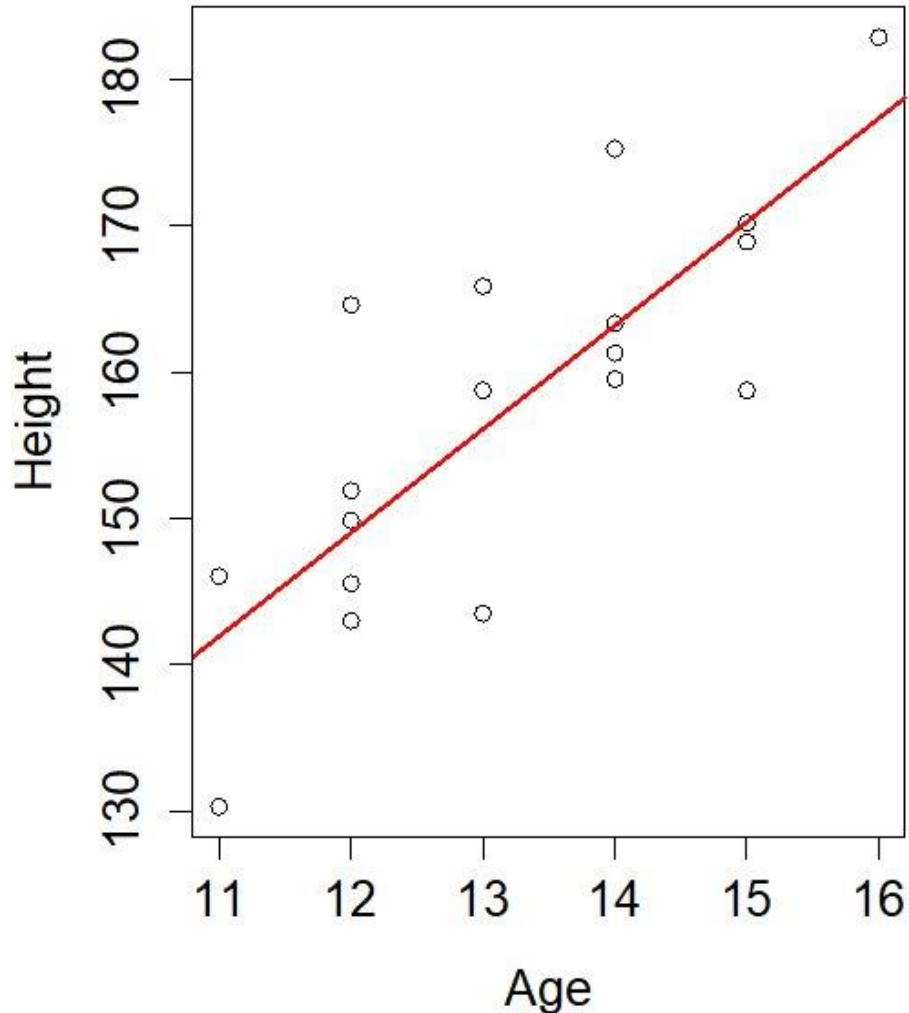
Coefficients:

(Intercept)	Age
64.069	7.079

→ **Height $\approx$ 64.1 + 7.1 $\times$ Age** (in cm, age in years)

Representation of the fit (II)

```
plot (Height~Age,data=class_data)  
abline (model_height_age, col="red", lwd=2)
```

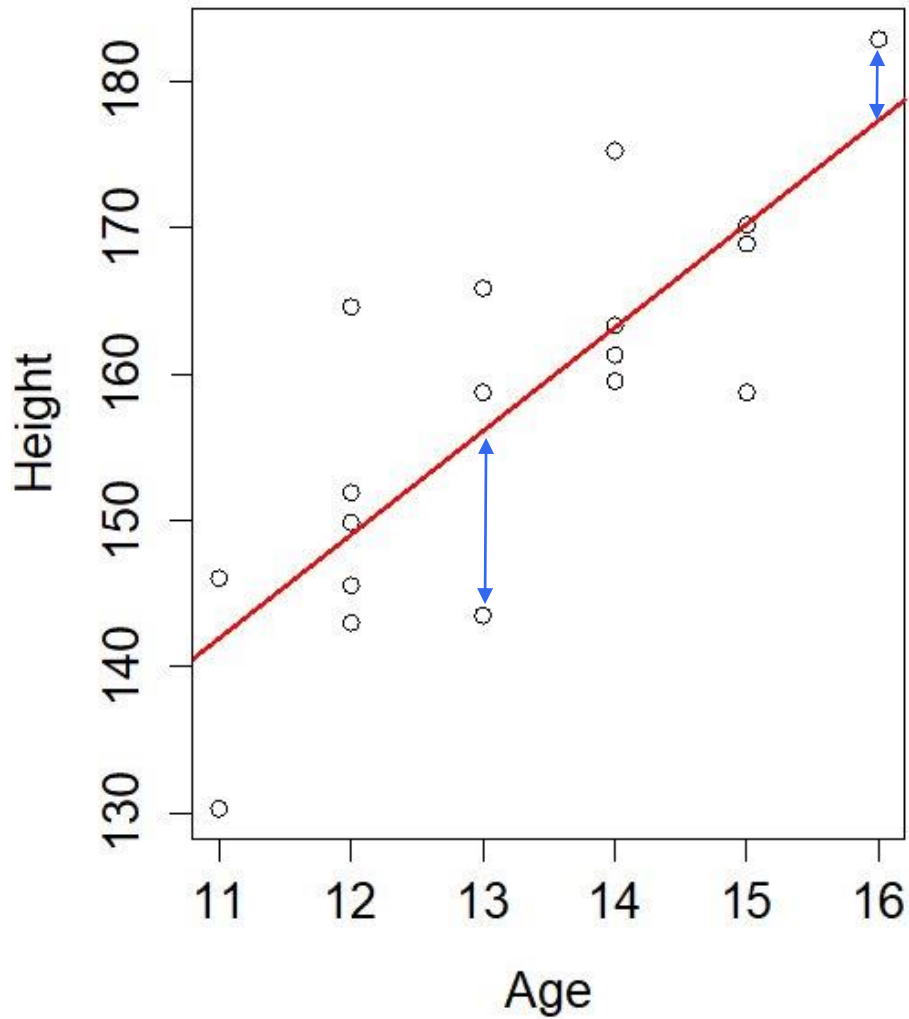


Coefficients = intercept and slope of the red line

```
coef(model_height_age)  
(Intercept)      Age  
  64.068667    7.079333
```

- **Slope:** each extra year of age adds on average about 7 cm of height
- **Intercept:** predicted height at age 0 (64 cm). Not meaningful here: there is no child younger than 11 in the data!

Representation of the fit (III)



Residuals: vertical distance between each point and the line (observed – predicted)

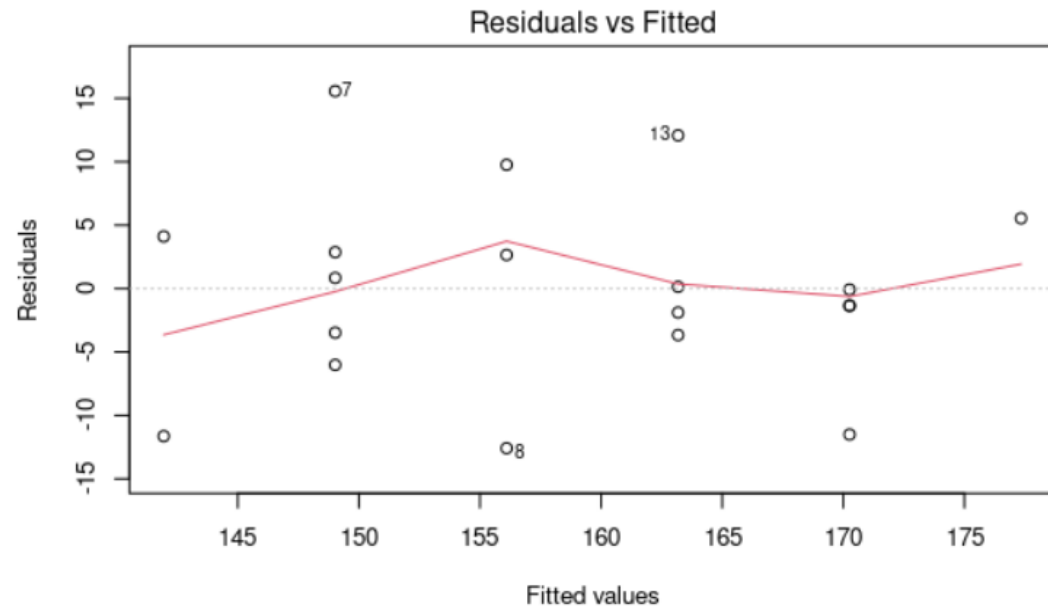
```
residuals(model_height_age)[1:4]
```

1	2	3	4
-11.639333	4.108667	-3.478667	2.871333

- Positive residual: the point is above the line
- Negative residual: the point is below the line
- A good model has small residuals with no visible pattern

Check model assumptions

```
par(mfrow=c(1,2))  
plot(model_height_age, which=1:2)  
# only the 2 most useful plots
```

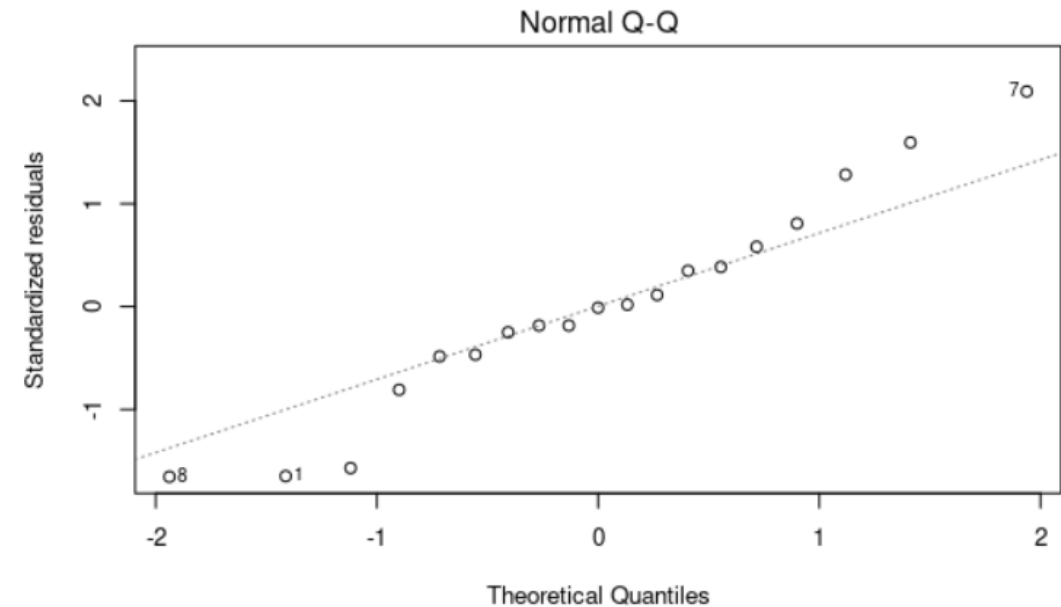


Residuals vs Fitted: are the residuals centred on 0?

- Good: points scattered randomly around the dotted line at 0, no clear shape
- Problem: a curve (the relation is not a straight line) or a funnel (spread increases)

More details on these plots:

<https://library.virginia.edu/data/articles/diagnostic-plots>



Normal Q-Q: are the residuals normally distributed?

- Good: points lie roughly on the dashed line
- Here: OK, small deviations at the ends are normal with only 19 children

Model summary

```
summary(model_height_age)
Call:
lm(formula = Height ~ Age, data = class_data)

Residuals:
    Min       1Q   Median       3Q      Max
-12.5900  -3.5730  -0.0787   3.4900  15.5713

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   64.069     16.565   3.868  0.00124 **
Age           7.079       1.237   5.724 2.48e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 7.832 on 17 degrees of freedom
Multiple R-squared:  0.6584, Adjusted R-squared:  0.6383
F-statistic: 32.77 on 1 and 17 DF, p-value: 2.48e-05
```

What to look at

- **Age, Estimate** = 7.079: the slope, height increases by about 7 cm per year
- **Age, Pr(>|t|)** = 2.48e-05: p-value testing whether the slope is different from 0. Very small → height is significantly associated with age
- **R-squared** = 0.66: age explains about 66% of the variation in height

You can ignore the rest of the output for now.

Let's practice - 11

The data set "Pima" comes from a study on diabetes in women of Pima Indian heritage.

We are using a subset (Pima.tr).

1. Load the package MASS using `library()`. Load the dataset Pima.tr using `data(Pima.tr)`. Use `?Pima.tr` to get an idea which variables it contains.
2. Hypothesis: Blood glucose level (glu) is associated with diastolic blood pressure (bp). Run a linear model to test the hypothesis.
3. Visualize the fit with a scatter plot and a trend line.
4. Check the assumptions of the model graphically with `plot()`.

EXTRA TASKS:

Try another explanatory variable, e.g. body mass index (bmi). Which of the two models explains more of the variation in glucose (R-squared)?

The next steps

- **R manuals:** <http://cran.r-project.org/manuals.html>
- **free course material:** <https://glittr.org>
- **STHDA (Statistical Tools for High Throughput Data Analysis) free tutorials:** <http://www.sthda.com/english/>
- **Stackoverflow** documentation, resources and user forum: <http://stackoverflow.com/tags/r/info>
- **Rseek** - search engine on numerous online R resources: <http://www.rseek.org>

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- Robert STOJNIĆ and Ian ROBERTS
- Jenny DRNEVICH

<http://www.sib.swiss/training>

Any questions? Contact training@sib.swiss

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